

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Mid-Semester Examination, 2025-26 (NEP)

DEPARTMENT OF PHYSICS**Paper Code: PHYMN41 / PHYMJ41 : Electromagnetic Theory & Relativistic Mechanics**

(Mid-Semester Examination — Batch P2)

Time: 1 Hour

Maximum Marks: 20

Instructions: Answer all questions from Section A and Section B as indicated. Figures to the right indicate full marks.

SECTION A**Question 1.** Answer all parts:

- (a) State divergence theorem. Give physical interpretation of the divergence theorem. [4]
- (b) Draw electric field lines produced by two equal and opposite point charges separated by some distance. [2]
- (c) Find the electric field a distance z above the center of a circular loop of radius r which carries a uniform line charge λ . [4]

SECTION B

- 2. A Michelson interferometer has arm length $L = 11$ cm, wavelength $\lambda = 500$ nm, and earth's velocity $v = 3 \times 10^4$ m/s. Calculate the expected fringe shift. Further, explain the negative (null) result obtained in Michelson-Morley experiment. [3]
- 3. Derive the Lorentz transformation equation starting from Einstein's postulates. [4]

OR

A rocket is chasing an enemy spaceship moving in the same direction. The enemy spaceship moves with a velocity of $0.6c$ relative to earth. The rocket moves with a velocity $0.8c$ relative to earth. Calculate the velocity of the rocket relative to the enemy spaceship using the relativistic velocity addition formula. [4]

- 4. Discuss the concept of zero rest mass of a photon and explain its implications in relativity. [3]
